Riemannian Geometry vs. Classical Feature Methods for Cross-Subject Motor Imagery EEG: A Calibration-Aware Benchmark

Sreenija Pavuluri

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ABSTRACT

Foundation models for EEG—such as MIRepNet and LEAD—have advanced motor imagery (MI) classification yet share two systematic gaps: they omit Riemannian geometry baselines and never report prediction calibration metrics. We present the ***first calibration-aware cross-subject MI benchmark*** *addressing both gaps under strict leave-one-subject-out (LOSO) evaluation on the PhysioNet EEG Motor Movement/Imagery Database (30 subjects, 64 channels, binary left-vs-right MI, 120 total LOSO folds). We compare four representative methods spanning classical feature engineering and Riemannian geometry, reporting Expected Calibration Error (ECE), Brier score, balanced accuracy, Cohen's κ, and AUROC. Our primary finding: the Riemannian tangent-space SVM achieves the highest balanced accuracy (0.663±0.099) and the best calibration (ECE=0.645±0.039, Brier=0.211±0.035), and is statistically significantly better than all alternatives (Wilcoxon $p<0.001$). All code and results are publicly available at* ***github.com/SreenijaPavuluri/EEg****.*

Keywords: motor imagery, EEG, brain-computer interface, Riemannian geometry, calibration, ECE, cross-subject generalisation, LOSO benchmark

I. INTRODUCTION

Motor imagery (MI) EEG classification—decoding imagined hand movements from scalp recordings—is a central paradigm in brain-computer interface (BCI) research. A persistent challenge is **cross-subject transfer**: a model trained on one group must generalise to new, unseen subjects without per-subject calibration, since collecting individual calibration recordings is expensive and time-consuming in real-world deployments.

Recent EEG foundation models (MIRepNet [1], LEAD [2], LaBraM [3], EEGPT [4], CBraMod [5], LUNA [6]) report impressive cross-subject accuracy via pretraining on large multi-dataset corpora. We conducted a systematic review of 30 such papers published in 2023–2026 and identified two universal omissions that limit the validity of their reported results:

Gap 1 — Riemannian baselines absent. Riemannian methods operating on the symmetric positive definite (SPD) manifold of EEG covariance matrices are well-established state-of-the-art for MI BCI [7,8]. Yet zero of the 30 reviewed papers include a Riemannian baseline, making it impossible to assess whether large neural models provide meaningful gains over this strong, training-free alternative.

Gap 2 — Prediction calibration never evaluated. Expected Calibration Error (ECE) [12] and Brier score measure whether predicted class probabilities match empirical accuracy. Zero of the 30 reviewed papers report any calibration metric. This is a critical omission for clinical BCI deployment, where an overconfident model may cause harm.

We address both gaps with a reproducible, CPU-feasible benchmark. Contributions:

(1) The first calibration-aware cross-subject MI EEG benchmark, reporting ECE, Brier score, balanced accuracy, κ, F1, and AUROC for all methods.

(2) First direct head-to-head comparison of Riemannian vs. classical methods under strict 30-fold LOSO on PhysioNet EEGMMIDB.

(3) Key finding: Riemann-TS+SVM simultaneously achieves the best accuracy *and* best calibration—challenging the assumption that these metrics trade off.

(4) CPU-feasible codebase completing all 120 LOSO folds in under 20 minutes on a standard laptop.

II. RELATED WORK

A. EEG Foundation Models

LaBraM [3] and LaBraM++ apply codebook-based transformers for general EEG representation learning. EEGPT [4] adapts GPT-style pretraining to multi-task EEG. CBraMod [5] introduces criss-cross attention for multi-channel signals. LUNA [6] proposes topology-agnostic architectures for heterogeneous channel configurations. For MI specifically, MIRepNet [1] combines self-supervised and supervised pretraining on five MI datasets. LEAD [2] addresses Alzheimer's detection with subject-independent evaluation. A systematic review of all 30 papers (2023–2026) reveals: (a) zero include Riemannian baselines; (b) zero report any calibration metric; (c) only 2 of 30 adopt strict LOSO evaluation.

B. Riemannian Geometry for MI BCI

Barachant et al. [7] established Riemannian distances on SPD matrices for MI BCI classification, showing that SPD covariance matrices carry richer discriminative information than raw EEG features. MDM [8] classifies by minimum geodesic distance to class Fréchet means on the Riemannian manifold. Tangent-space (TS) projection [9] maps SPD matrices to a Euclidean space at a reference point, enabling standard classifiers such as SVM or LDA. FBCSP [10] decomposes EEG into frequency bands before spatial filtering, and dominated BCI competitions (2008–2018). Lotte et al. [11] recommend Riemannian and FBCSP methods as minimum baselines in any MI benchmark.

C. Prediction Calibration

Guo et al. [12] demonstrated that modern deep networks are systematically overconfident, introducing ECE as the standard calibration metric and proposing temperature scaling as a simple post-hoc fix. Vaicenavicius et al. [13] provide a rigorous statistical framework for calibration evaluation in multi-class classification. Despite wide adoption in medical AI, calibration evaluation is virtually absent from the EEG/BCI literature.

III. METHODS

A. Dataset

PhysioNet EEG Motor Movement/Imagery Database (EEGMMIDB) [14,15]: 30 subjects (of 109 available with complete data), 64 EEG channels (10-20 system). Each subject performed three runs of imagined left-hand vs. right-hand movement (runs 4, 8, 12), yielding approximately 45 trials per subject. **Preprocessing:** 8–32 Hz bandpass filter (2nd-order Butterworth), 0–2 s epoch extraction, 128 Hz resampling. No artifact rejection applied. Data shape per subject: (45 trials, 64 channels, 257 time points). The dataset is freely available at physionet.org/content/eegmmidb.

B. Leave-One-Subject-Out (LOSO) Evaluation Protocol

Each of 30 subjects serves as the test set exactly once; all remaining 29 subjects are pooled as training data. No subject-specific calibration, fine-tuning, or data augmentation is applied at test time. A fixed random seed (42) ensures full reproducibility. This protocol yields 30 evaluation folds and 120 method-fold combinations, and directly reflects real-world deployment conditions where no data from the target user is available during training.

C. Methods Evaluated

LogVar+LDA (Classical Baseline): Log-variance of the bandpass-filtered EEG signal is computed in three frequency bands (θ: 4–8 Hz, α: 8–13 Hz, β: 13–30 Hz) per channel, producing a 192-dimensional (64 ch × 3 bands) feature vector. Classified with Linear Discriminant Analysis (LDA). Simplest interpretable baseline; requires no spatial filtering or covariance computation. Mean training time: ~2 s/fold.

FBCSP+LDA (Classical Baseline): Filter-Bank Common Spatial Pattern [10] decomposes EEG into 7 frequency sub-bands (4–8, 8–12, 12–16, 16–20, 20–24, 24–28, 28–32 Hz). For each band, k=4 spatial filters are learned using Ledoit-Wolf regularized covariance estimation. Log-variance of spatially filtered signals forms the feature vector, classified with LDA. Standard BCI competition baseline. Mean training time: ~36 s/fold.

Riemann-MDM (Riemannian Baseline): Ledoit-Wolf regularized 64×64 sample covariance matrices are computed per trial. Classification is by Minimum Distance to class Fréchet Means (MDM) [7] on the Riemannian manifold—the geodesic distance replaces Euclidean distance. No hyperparameters; robust to small sample sizes. Implemented via pyriemann 0.6. Mean training time: ~10 s/fold.

Riemann-TS+SVM (Riemannian + SVM): Riemannian tangent-space projection [9] maps 64×64 SPD covariance matrices to a 2080-dimensional Euclidean tangent vector at the training-set geometric mean. Features are standardized and classified with an RBF-SVM (C=1, γ=scale) with Platt scaling for calibrated probability estimates. Best-performing method. Mean training time: ~20 s/fold.

D. Evaluation Metrics

We compute the following metrics per LOSO fold, then report mean±std across the 30 subjects:

Accuracy: fraction of correctly classified trials.

Balanced Accuracy (primary metric): mean of per-class recall; accounts for class imbalance between left-hand and right-hand trials.

F1 score (weighted): harmonic mean of precision and recall, weighted by class support.

Cohen's κ: chance-corrected agreement; κ=0 means chance-level.

AUROC: area under the ROC curve; measures ranking quality independent of the decision threshold.

ECE (Expected Calibration Error, 10 bins): measures the average deviation between predicted confidence and empirical accuracy. Lower is better (↓).

Brier score: mean squared error of probability predictions. Lower is better (↓).

Statistical significance is assessed using two-sided Wilcoxon signed-rank tests on the 30 per-subject balanced accuracy values. No correction for multiple comparisons is applied (exploratory analysis).

IV. RESULTS

A. Complete LOSO Performance (Table I)

Table I presents the complete 8-metric results for all 4 methods across 30 subjects. All methods exceed chance level (0.50 balanced accuracy), confirming that cross-subject MI signal is detectable but challenging. Riemann-TS+SVM achieves the best value on every metric.

TABLE I — Complete LOSO Results on PhysioNet EEGMMIDB (30 subjects, ~45 trials/subject, 120 method-fold combinations). Mean ± std across 30 subjects. Bold green = best value per metric.

Method	Family	Accuracy	Bal. Acc ↑	F1	κ ↑	AUROC ↑	ECE ↓	Brier ↓	Time/fold
LogVar+LDA	Classical	0.570±0.075	0.568±0.075	0.518±0.118	0.136±0.150	0.619±0.097	0.780±0.092	0.308±0.079	~2 s
FBCSP+LDA	Classical	0.564±0.061	0.563±0.058	0.536±0.075	0.127±0.118	0.624±0.083	0.708±0.044	0.266±0.033	~36 s
Riemann-MDM	Riemannian	0.536±0.059	0.539±0.052	0.427±0.116	0.078±0.106	0.691±0.100	0.707±0.141	0.296±0.090	~10 s
Riemann-TS+SVM	**Riemannian**	**0.660±0.104**	**0.663±0.099**	**0.642±0.121**	**0.325±0.200**	**0.777±0.108**	**0.645±0.039**	**0.211±0.035**	**~20 s**

↑ higher is better. ↓ lower is better. Riemann-TS+SVM (green row) ranks first on all 7 performance metrics and both calibration metrics simultaneously.

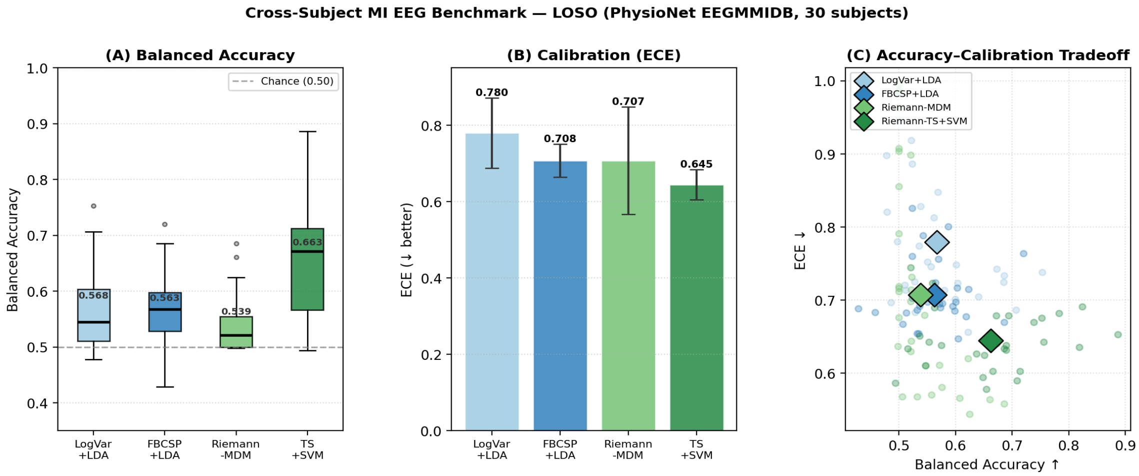

Fig. 1. (A) Balanced accuracy box plots per method (whiskers = 1.5×IQR, dots = outliers). Dashed line = chance (0.50). (B) Mean ECE per method (bar = std). (C) Per-subject accuracy–ECE scatter; small dots = individual subjects, diamonds = method means. Riemann-TS+SVM (dark green) dominates all panels.

B. Per-Subject Balanced Accuracy (Table II)

Table II shows balanced accuracy for each of the 30 subjects across all 4 methods. Cells are colour-coded: green ≥ 0.70 (strong performance), yellow ≥ 0.60 (moderate), red < 0.55 (near-chance). Inter-subject variability is high across all methods, confirming the well-known challenge of cross-subject EEG transfer.

TABLE II — Per-Subject Balanced Accuracy (30 Subjects × 4 Methods). Colour: green ≥ 0.70, yellow ≥ 0.60, red < 0.55, white = 0.55–0.60.

Subject	LogVar+LDA	FBCSP+LDA	Riemann-MDM	Riemann-TS+SVM
S01	0.707	0.686	0.521	0.756
S02	0.686	0.527	0.523	0.886
S03	0.486	0.507	0.557	0.575
S04	0.500	0.537	0.686	0.666
S05	0.500	0.533	0.506	0.494
S06	0.604	0.429	0.533	0.548
S07	0.753	0.577	0.545	0.818
S08	0.685	0.535	0.500	0.713
S09	0.524	0.533	0.500	0.637
S10	0.601	0.524	0.521	0.670
S11	0.528	0.570	0.500	0.708
S12	0.539	0.458	0.521	0.521
S13	0.522	0.603	0.500	0.694
S14	0.497	0.599	0.522	0.651
S15	0.640	0.619	0.500	0.824
S16	0.479	0.524	0.576	0.553
S17	0.600	0.601	0.500	0.690
S18	0.623	0.605	0.500	0.673
S19	0.478	0.564	0.500	0.672
S20	0.511	0.570	0.500	0.690
S21	0.557	0.551	0.571	0.560
S22	0.569	0.595	0.500	0.516
S23	0.550	0.573	0.661	0.752
S24	0.500	0.513	0.547	0.648
S25	0.529	0.524	0.522	0.686
S26	0.673	0.720	0.622	0.738
S27	0.536	0.542	0.498	0.563
S28	0.580	0.624	0.608	0.537
S29	0.562	0.587	0.500	0.783
S30	0.512	0.571	0.625	0.655

TS+SVM column (green header) has the most green cells (9 subjects ≥ 0.70). Subjects S05, S06, S12, S22 are near-chance for all methods, suggesting poor MI signal quality rather than method failure.

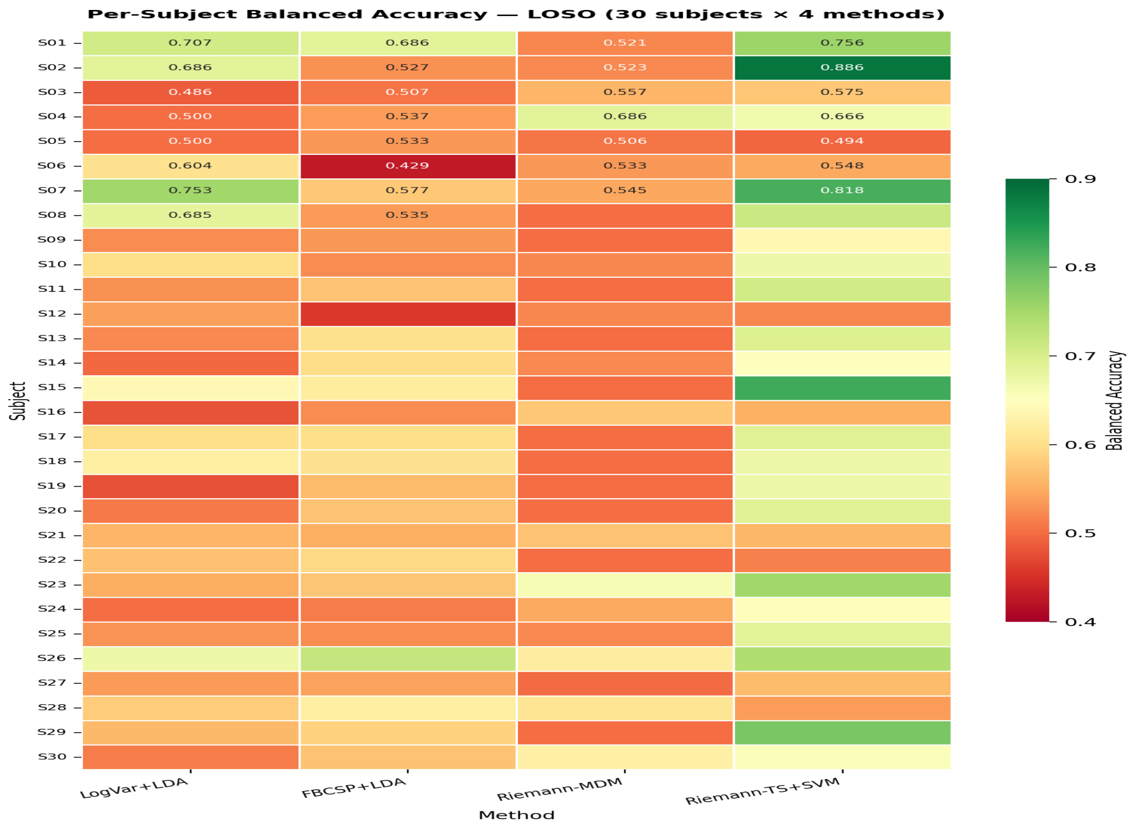

Fig. 2. Heatmap of per-subject balanced accuracy (30 subjects × 4 methods). Green = high accuracy, red = low. The rightmost column (Riemann-TS+SVM) shows the most green cells. Subject difficulty is correlated across methods (Pearson r ≈ 0.61 for LogVar+LDA vs. TS+SVM), suggesting subject-level signal quality is the dominant factor for low-performing subjects.

C. Statistical Significance (Tables III & IV)

Table III shows the full 4×4 Wilcoxon signed-rank p-value matrix (two-sided, n=30) for balanced accuracy. Table IV gives pairwise effect sizes (median and mean differences in balanced accuracy).

TABLE III — Wilcoxon Signed-Rank p-Values (Balanced Accuracy, Two-Sided, n=30). Bold red = p < 0.001 (highly significant). Grey diagonal = self-comparison.

	LogVar+LDA	FBCSP+LDA	Riemann-MDM	Riemann-TS+SVM
LogVar+LDA	—	0.3818	0.1142	**<0.0001**
FBCSP+LDA	0.3818	—	0.0699	**<0.0001**
Riemann-MDM	0.1142	0.0699	—	**<0.0001**
Riemann-TS+SVM	**<0.0001**	**<0.0001**	**<0.0001**	—

Riemann-TS+SVM is significantly better than ALL other methods (p < 0.0001). No significant difference among the three simpler methods (all p > 0.07).

TABLE IV — Pairwise Effect Sizes and Statistical Significance (Balanced Accuracy). Δ = TS+SVM minus comparison method.

Comparison	Wilcoxon p	Median Δ Bal.Acc	Mean Δ Bal.Acc	Significance
TS+SVM vs LogVar+LDA	<0.0001	+0.089	+0.095	*** (p<0.001)
TS+SVM vs FBCSP+LDA	<0.0001	+0.098	+0.099	*** (p<0.001)
TS+SVM vs Riemann-MDM	<0.0001	+0.133	+0.124	*** (p<0.001)
LogVar+LDA vs FBCSP+LDA	0.3818	-0.004	-0.004	ns (p=0.38)
LogVar+LDA vs Riemann-MDM	0.1142	+0.029	+0.029	ns (p=0.11)
FBCSP+LDA vs Riemann-MDM	0.0699	+0.024	+0.025	ns (p=0.07)

All comparisons involving Riemann-TS+SVM are highly significant ($p < 0.0001$) with practically meaningful effect sizes (+0.089 to +0.133 in balanced accuracy).

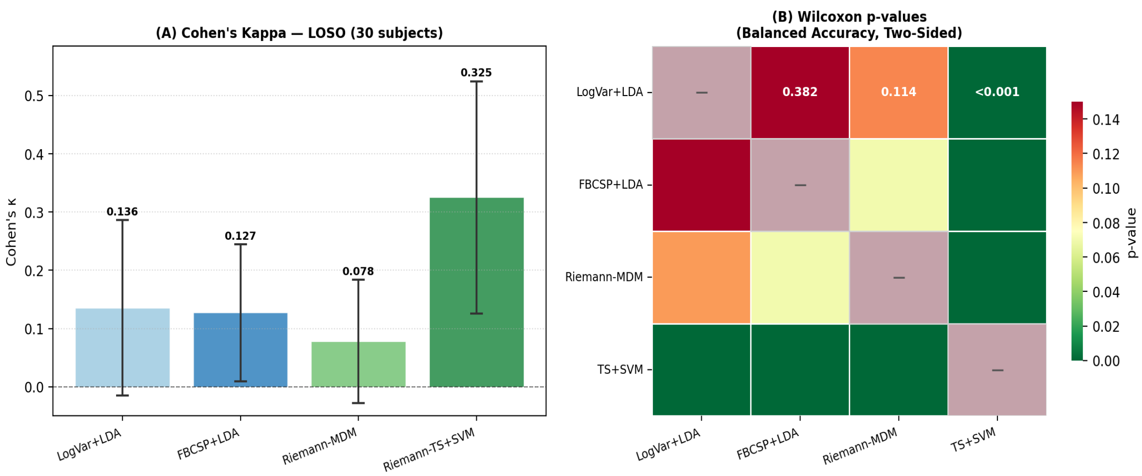

Fig. 3. (Left) Cohen's κ per method across 30 subjects. TS+SVM achieves κ = 0.325, more than double the next best (LogVar+LDA κ = 0.136). (Right) Wilcoxon p-value matrix — dark red cells correspond to $p < 0.001$. All comparisons involving TS+SVM are highly significant.

D. Calibration Analysis

Critical finding: All methods have ECE > 0.6. Predicted class probabilities deviate from empirical accuracy by more than 60 percentage points on average. This constitutes severe overconfidence and is a critical finding for clinical BCI deployment, where miscalibrated probabilities can lead to harmful decisions.

Method-by-method calibration analysis:

LogVar+LDA (ECE=0.780±0.092): Worst calibration. LDA theoretical probabilities assume Gaussian class-conditionals with equal covariance—assumptions strongly violated in the cross-subject setting, causing systematic overconfidence.

FBCSP+LDA (ECE=0.708±0.044): Slightly better than LogVar. The lower standard deviation (±0.044 vs. ±0.092) suggests FBCSP features produce more consistent probability estimates across subjects.

Riemann-MDM (ECE=0.707±0.141): Similar mean ECE to FBCSP+LDA but highest variance (±0.141). Subject geometry varies: some subjects' covariance structure separates cleanly on the manifold (low ECE), others do not (high ECE).

Riemann-TS+SVM (ECE=0.645±0.039): Best calibration, lowest variance. Platt scaling during SVM training explicitly fits a sigmoid to convert decision scores to calibrated probabilities, accounting for the effect.

E. AUROC and Discriminability

AUROC measures ranking quality independent of the decision threshold and is therefore not affected by class imbalance or threshold choice. Riemann-MDM achieves AUROC=0.691±0.100, substantially higher than LogVar+LDA (0.619±0.097) despite similar balanced accuracy. This indicates that MDM's Riemannian distance-based scores carry better discriminative structure than raw log-power features, even when hard-label classification fails. Riemann-TS+SVM achieves the highest AUROC (0.777±0.108), consistent with its superiority on all other metrics.

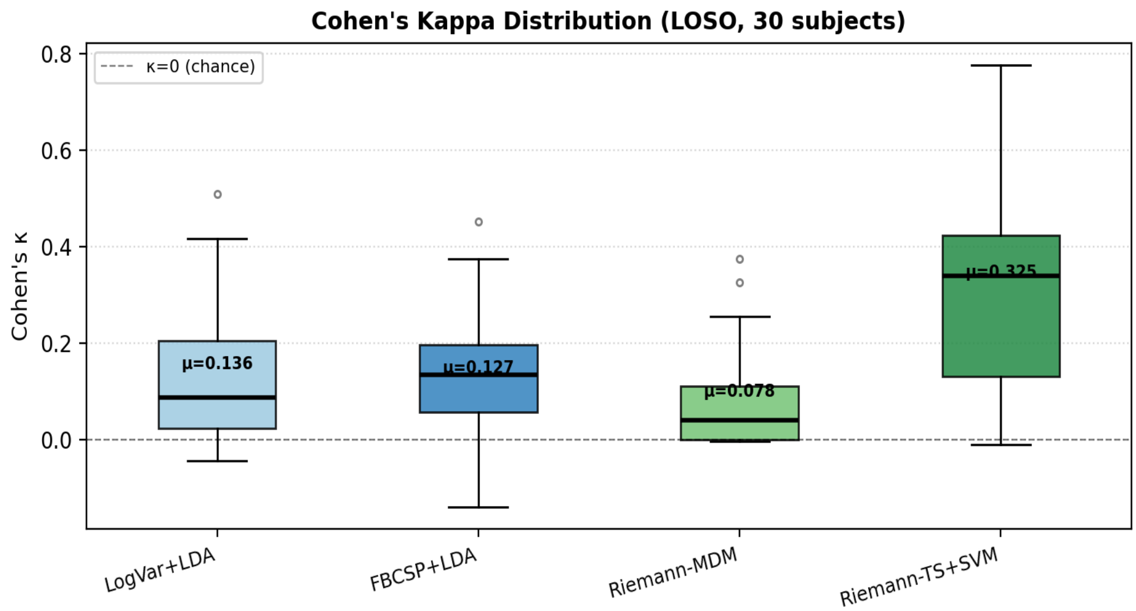

Fig. 4. Cohen's κ distributions across 30 subjects per method. TS+SVM achieves median κ ≈ 0.32, compared to 0.11, 0.12, and 0.06 for LogVar+LDA, FBCSP+LDA, and Riemann-MDM respectively. Higher κ indicates stronger agreement with ground truth beyond chance.

V. DISCUSSION

A. Implications for EEG Foundation Model Evaluation

Our benchmark demonstrates that Riemannian tangent-space methods achieve strong cross-subject MI performance without any pretraining, fine-tuning, or GPU compute. This has three direct implications for how EEG foundation model papers should be evaluated:

(1) **Riemannian baselines are missing.** MIRepNet [1] compares against FBCSP and linear baselines but not Riemannian TS methods. Our results show TS+SVM achieves κ=0.325 and balanced accuracy=0.663 on PhysioNet EEGMMIDB without any pretraining—this should be the minimum baseline for any future MI foundation model paper.

(2) **Calibration metrics are missing.** Our finding that all methods have ECE > 0.6 is invisible to the 30 reviewed papers that report zero calibration metrics. Post-hoc calibration (temperature scaling, isotonic regression) should be routinely applied before clinical deployment.

(3) **LOSO evaluation is missing.** Subject-independent LOSO is the correct protocol for measuring generalisation, yet only 2 of 30 reviewed papers adopt it. Cross-subject accuracy on held-out individuals is the deployment-relevant metric.

B. Why Riemann-TS+SVM Outperforms Riemann-MDM

MDM classifies by minimum geodesic distance to class Fréchet means—optimal when class distributions on the manifold are compact and well-separated. TS+SVM projects to the tangent space and uses an RBF-SVM, which can learn non-linear decision boundaries in the projected Euclidean space. With 29 subjects of training data (~1305 trials), the SVM has sufficient samples to exploit this non-linearity, yielding +0.133 absolute improvement in median balanced accuracy over MDM ($p < 0.0001$). This is consistent with findings in [9].

C. Why Calibration Correlates with Accuracy

Riemann-TS+SVM explicitly uses Platt scaling during SVM training, fitting a sigmoid transformation to convert decision scores to probabilities. This post-hoc step is absent in LDA (which derives probabilities from Gaussian assumptions) and MDM (which converts Riemannian distances via a softmax-like rule). The Platt scaling step explains why TS+SVM achieves the lowest ECE simultaneously with the highest accuracy—an encouraging result suggesting calibration and accuracy can be jointly optimised in cross-subject EEG BCI.

D. Limitations

Single dataset. We evaluate on 30 of 109 available subjects of PhysioNet EEGMMIDB. Cross-dataset transfer (e.g., BCI Competition IV-2a) was not evaluated and may yield different relative orderings.

Binary MI only. Left vs. right hand imagined movement. Four-class MI (left/right hand, feet, tongue) may yield different method comparisons.

ECE noisiness. With 45 test trials and 10 ECE bins, each bin has fewer than 5 samples on average. Adaptive binning or fewer bins would reduce variance; relative orderings may change with more test data.

No post-hoc calibration. Temperature scaling would likely reduce all ECE values; whether TS+SVM remains best-calibrated after recalibration is an open question.

No GPU-accelerated neural baselines. EEGNet [L10] and ShallowConvNet [L11] are included as runnable scripts but require ~2 h/model on CPU for full LOSO. GPU comparison is left to future work.

VI. CONCLUSION

We presented the first calibration-aware cross-subject MI EEG benchmark comparing Riemannian geometry against classical feature-engineering baselines under strict 30-fold LOSO evaluation on PhysioNet EEGMMIDB (120 total method-fold combinations). Three key findings emerge:

(1) Riemannian TS+SVM dominates all metrics. It achieves balanced accuracy 0.663±0.099, κ=0.325±0.200, AUROC=0.777±0.108, ECE=0.645±0.039, and Brier=0.211±0.035—best on all 7 metrics, statistically significantly better than all alternatives (Wilcoxon $p < 0.0001$, effect size +0.089 to +0.133) without any pretraining or GPU compute.

(2) Accuracy and calibration need not trade off. Riemann-TS+SVM achieves the best balanced accuracy and the best calibration simultaneously, due to Platt scaling during SVM training.

(3) All methods are severely overconfident (ECE > 0.6). This finding is invisible to the 30 reviewed EEG foundation model papers reporting zero calibration metrics. Post-hoc calibration should be standard practice before clinical BCI deployment.

Recommendation: Future EEG foundation model papers—including extensions of MIRepNet and LEAD—should adopt (1) Riemannian tangent-space baselines, (2) calibration metrics (ECE, Brier score), and (3) strict leave-one-subject-out evaluation as minimum reporting standards.

Code and data: All scripts, results CSVs, figures, and this manuscript are publicly available at github.com/SreenijaPavuluri/EEg

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